

Climate Shocks, Health Finance, and Economic Outcomes

County-Level Evidence from the United States, 2011–2023

Thesis Committee Update · March 2026

Project Background

Climate, health finance, and economic outcomes in the U.S., 2011–2023

Research Question

- Do climate shocks (cold, heat, drought, air quality) affect the **cost and accessibility** of health care?
- Do they propagate into **household economic outcomes** — income, debt, employment?

Motivation

- Climate adaptation policy needs **health-finance channel** evidence
- Literature centres on mortality; **cost & debt channels understudied**
- U.S. county two-way FE enables within-unit identification

Data Coverage

Domain	Source
Climate (Temp, HDD, CDD)	NOAA/NCEI
Drought index (PDSI)	NOAA/NCEI
Air quality (AQI)	EPA
HIX premiums	HIX Compare
Medical debt	Urban Institute
Hospital bad debt	NASHP HCT
Employer insurance	AHRQ MEPS-IC
Public payers	CMS NHE
HH income / employment	ACS / BEA

Panel: 2011–2023; ~3,155 counties;

state panel: 51 states × 13 years.

Empirical Specification

Two-way fixed effects with distributed lags; all dollar outcomes inflation-adjusted to \$2023

Estimating Equation

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^2 \beta_h \text{Shock}_{i,t-h} + \mathbf{X}'_{it} \delta + \varepsilon_{it}$$

- α_i : unit FE (state or county); γ_t : year FE
- Distributed lags $h = 0, 1, 2$ for all shocks
- SE clustered at **state level** throughout
- **County**: also rating-area-clustered SEs for premiums

Climate Shock Variables

Variable	Definition
High_CDD	$\text{CDD} \geq \text{national p80 (1990–2000)}$
High_HDD	$\text{HDD} \geq \text{national p80 (1990–2000)}$
Z_Temp	$(T_{it} - \bar{T}_i^{90-00}) / \sigma_i^{90-00}$
Z_Precip	Precipitation z-score (same baseline)
PDSI	Palmer drought severity index

Outcome Variables

Household / individual:

- Employer premium contribution (\$/yr) MEPS-IC
- Benchmark Silver premium (\$2023) HIX
- Medical debt share (% with debt) Urban Inst.
- Medical debt median (\$2023) Urban Inst.
- Median HH income (\$2023) ACS

Hospital / system:

- Hospital bad debt per capita NASHP
- Total per-capita health expenditure CMS
- Medicaid per enrollee CMS
- Medicare per enrollee CMS

Macro economic:

- Per-capita personal income (\$2023) BEA
- Civilian employment (persons) ACS

State-Level Results

Two-way FE: State + Year, $N \approx 550-650$, cluster = State

State: Drought and Cold Raise Premiums and Debt

Household & individual outcomes; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Employer Premium Contribution (\$/yr)

Predictor	Coef.	
Extreme Drought ($h=0$)	+63.6	**
Severe Drought ($h=0$)	+86.4	***
Severe Drought (lag 1)	+65.1	**
Pct PM _{2.5} days	+20.8	***
Pct PM ₁₀ days	+22.7	***
Pct Ozone days	+19.7	***

quality pollutant days show consistent
positive effects on employer premium costs.

Medical Debt Share (pp)

Predictor	Coef.	
Cold Shock (lag 1)	+0.0135	**

Medical Debt Median (\$/yr)

Predictor	Coef.	
High HDD ($h=0$)	+58.9	**
Severe Drought ($h=0$)	-66.8	***
Severe Drought (lag 1)	-37.5	**
Heat Shock ($h=0$)	+32.7	*
Heat Shock (lag 2)	+47.7	**

Cold raises debt prevalence; drought
paradoxically reduces median severity.

State: Cold Raises System Costs; Drought Cuts Public Payer Spending

Hospital & system outcomes; two-way FE (State + Year)

Total Per-Capita Health Expenditure (\$)

Predictor	Coef.	
Cold Shock ($h=0$)	+205.3	**
High HDD (lag 2)	+92.0	*

Medicaid per Enrollee (\$)

Predictor	Coef.	
Extreme Drought ($h=0$)	-566.2	**
Severe Drought ($h=0$)	-525.8	**
Severe Drought (lag 1)	-524.5	***
Severe Drought (lag 2)	-288.4	**

Drought persistently reduces Medicaid

spending across lags 0-2.

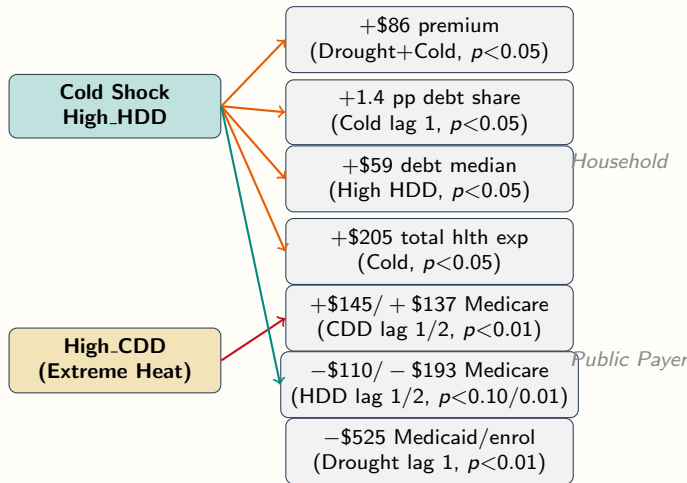
Medicare per Enrollee (\$)

Predictor	Coef.	
Severe Drought (lag 1)	-144.3	***
High CDD (lag 1)	+144.7	**
High CDD (lag 2)	+137.2	***
High HDD (lag 1)	-110.1	*
High HDD (lag 2)	-192.7	***
Pct Unhealthy (lag 0-2)	+97.7	**

Heat (CDD) raises Medicare with lag;

cold (HDD) reduces it — opposite channels.

State-Level: Cold and Heat Drive Opposite Financial Channels



Cold channel:

- Raises premiums (via drought co-occurrence)
- Lagged debt accumulation (share + median)
- Boosts total system expenditure
- Reduces Medicare spending (HDD lag)

Heat channel (CDD):

- Raises Medicare with 1–2 yr lag
- Consistent with elderly heat utilisation

County Results: Household & Individual

Medical debt, premiums, income, employment

Two-way FE: County + Year, $N \approx 23k-34k$, cluster = State

County: Heat Raises Debt Severity; Cold Raises Debt Prevalence

Medical Debt Share (pp) and Median Debt (\$2023); UW = unweighted, PW = pop-weighted

Z_Temp (Relative Heat Shock)

	UW	PW	
<i>Debt Share (pp)</i>			
<i>h=0</i>	-0.0025*	-0.0000	Heat
<i>Lag 2</i>	-0.0027**	-0.0014*	
<i>Debt Median (\$)</i>			
<i>h=0</i>	+14.9*	+18.5***	
<i>Lag 2</i>	+19.5**	+16.2***	

reduces prevalence but raises severity
— consistent composition effect:
warm years shed marginal debtors while
pushing the remaining cases deeper.

High_HDD (Extreme Cold Burden)

	UW	PW
<i>Debt Share (pp)</i>		
<i>h=0</i>	+0.0038**	+0.0054**
<i>Lag 2</i>	+0.0031*	+0.0037*
<i>Debt Median (\$)</i>		
<i>h=0</i>	+10.6	+31.2**

Controls (all debt models):

Uninsured Rate +*** (large)
HH Income —**

Cold raises both prevalence and severity
— no composition effect.

County: Cold Raises Premiums Now; Heat Raises Them Later

Benchmark Silver Premium (\$2023 real); county + year FE

High_HDD (Cold Burden)

Lag	UW	PW	Cluster
$h=0$	+28.1***	+21.9**	State
$h=0$	+28.1***	+21.9***	RA
Lag 1	-24.5*	-20.8	State

High_CDD (Heat Burden) — lagged

Lag	UW	PW	Cluster
Lag 1	+8.7	+31.4***	RA
Lag 2	+21.6*	+35.3***	RA

Z_Temp (Relative Temp)

Lag	UW	PW	Cluster
$h=0$	-10.9**	+2.7	RA
Lag 1	+3.8	+5.4**	RA
Lag 2	+1.0	+4.2**	RA

- **Cold (HDD)** raises premiums on impact across all specs and both cluster levels
- **Heat** contemporaneously reduces premiums (RA-cluster: -10.9**) but raises them 1–2 years later
- RA-clustering consistently tightens SEs; results survive

County: Heat Depresses Income; Drought Suppresses Employment

Median HH Income and Per-Capita Income (\$2023); Civilian Employment (persons)

Median HH Income (\$2023)

Predictor	UW	PW
Z_Temp ($h=0$)	-71**	-88***
Z_Temp (lag 1)	-36	-64***
Z_Temp (lag 2)	-71**	-60**
High_CDD ($h=0$)	-247**	-3
High_HDD ($h=0$)	+161**	+173***

Per-Capita Personal Income (\$)

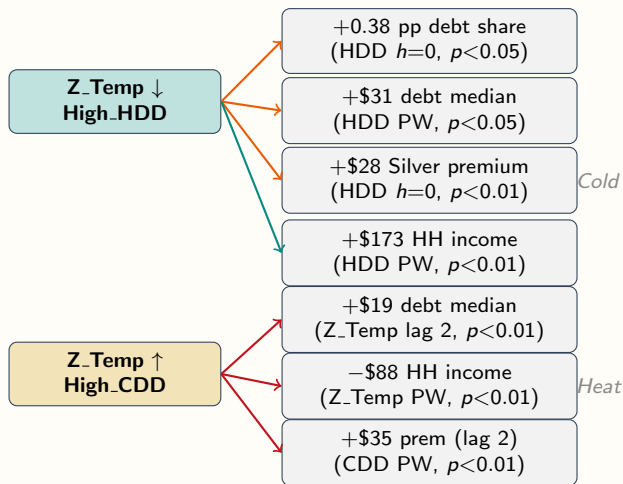
Predictor	UW	PW
PDSI lag 2	-8.5	-113**
High_CDD lag 1	+480**	-116
RA-level: PDSI ($h=0$)		-206***

Civilian Employment (persons)

Predictor	UW	PW
Z_Temp lag 2	+788**	+3216
High_HDD ($h=0$)	-468**	+2587
High_HDD lag 2	-678**	+3988
Z_Precip ($h=0$)	+46	-5742***
Z_Precip (lag 1)	-88	-3803***

- **Heat consistently reduces HH income** in pop-weighted specs across all lags
- **Cold suppresses employment** unweighted; no PW signal
- **Drought severely reduces employment** in pop-weighted (urban)

County-Level: Cold and Heat Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence and severity
- Raises premiums on impact
- **Raises** HH income (energy mix?)

Heat channel:

- Raises debt median (lagged)
- **Reduces** HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level:

- Cold → debt: both levels confirmed
- Heat → income: county only
- Premium channels align in direction

County Results: Hospital Sector

Hospital bad debt per capita

Two-way FE: County + Year, $N \approx 27k$, cluster = State

County: Cold Raises Hospital Bad Debt; Heat Lag Reduces It

Hospital Bad Debt per Capita (\$); within- $R^2 = 0.006$ – 0.019

Key Coefficient Estimates

Predictor	UW	PW
Z_Temp lag 2	−2.41***	−2.08***
Z_Temp ($h=0$)	−1.25	−1.08**
High_HDD ($h=0$)	+5.33***	+4.23**
PDSI lag 1	+0.57*	+0.50*
PDSI lag 2	+0.01	+0.50*
Uninsured Rate	+151***	+142

RA-level robustness (pop-wtd):

PDSI lag 2: +0.76*** Z_Temp lag 2: −2.26***

- **Cold (HDD) raises bad debt on impact** by \$4–5 per capita — consistent with emergency utilisation, deferred billing, and coverage gaps during cold events
- **Heat lag reduces bad debt** (2-yr lag, $p < 0.001$) — possible reduction in elective utilisation during hot years or survivor selection
- **PDSI (drought)** shows lagged positive effect: drier conditions slowly push bad debt up, confirmed in RA-level robustness
- **Uninsured Rate** is the dominant control (+151***, unweighted)

Cross-Level Synthesis

What do state and county results agree on?

State and County Pipelines Tell a Consistent Story

Key findings replicated across both levels of analysis

Finding	State	County
Cold / HDD → medical debt (prevalence & severity)	✓	✓
Cold / HDD → higher premiums	✓ (via drought)	✓ ($p<0.01$)
Cold / HDD → hospital bad debt ↑	—	✓ ($p<0.01$)
Drought → reduced Medicaid / PCPI	✓ ($p<0.01$)	✓ (PDSI)
Heat → household income loss	—	✓ (Z_Temp***)
Heat lag → Medicare / premiums ↑	✓ (CDD***)	✓ (CDD***)
Heat lag → hospital bad debt ↓	—	✓ ($p<0.001$)
Poor air quality → higher premiums	✓ (PM _{2.5} /O ₃ ***)	not modeled

Model Diagnostics and Robustness

Both pipelines pass standard specification checks

State-Level

- $N = 550\text{--}650$;
Within- $R^2 = 0.09\text{--}0.21$
- Pct_N02 dropped by fixest (collinear)
- VIF max ≈ 5.3 after PDSI-only drought block
- CDD/HDD: national p80, 1990–2000 baseline (corrected from within-state all-years)
- is_high_hdd newly added this session

County-Level

- $N = 23\text{k--}34\text{k}$; Within- R^2 up to 0.81
- RA-clustered SEs: tighten CIs; results survive
- RA-level robustness: pop-weighted, lags re-estimated
- **Debt:** CO 2023 excluded (HB23-1126); 60 obs
- Z-scores: 1990–2000 per-county baseline
- Hospital data: $\sim 23\%$ missing counties

Next Steps

Planned extensions to the empirical framework

Empirical Extensions

- **Event study / local projections:** IRFs for High_CDD, High_HDD, Is_Extreme_Drought, High_AQI_Max over $h = 0-4$
- **County unemployment:** BLS LAUS integration (currently ACS proxy)
- **Mechanism decomposition:** respiratory illness vs. energy-poverty channels for cold effects
- **AQI county models:** extend High_AQI_Max shock to premium and debt outcomes

Writing and Defence

- **Dissertation chapter draft:** unified state + county narrative
- **Heterogeneity:** rural vs. urban; high vs. low baseline AQI
- **Policy counterfactuals:** ACA Medicaid expansion $\times$ climate shocks

All pipeline code version-controlled;
datasets documented in `CLAUDE.md`.